

PROJECT DESIGN PHASE – II

Data Flow Diagram & User Stories

Team ID	SWTID-2026-5397
Team Size	5
Project Name	Import Data using Transform Maps (Spreadsheet)
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Project Overview

The project provides a structured method to import records from a spreadsheet into a target system using a Transform Map. The process validates the uploaded file, maps source columns to target fields, applies transformation rules, identifies existing records, creates or updates records, and produces an import result for the user.

Objectives

- Import spreadsheet data into the selected target table.
- Map spreadsheet columns to target fields accurately.
- Apply field transformations and validation rules.
- Prevent duplicate or invalid records where applicable.
- Provide import status, error information, and a completion summary.

Scope

The documentation covers the spreadsheet import workflow, source-to-target mapping, transformation, validation, record insertion/update, error handling, and user-facing import status.

1. Data Flow Diagram (DFD)

A Data Flow Diagram represents how data enters the system, how it is processed, where it is stored, and how results are returned to the user.

Data Administrator / User Uploads spreadsheet Selects target table	Import Set Receives uploaded file Reads source columns	Transform Map Maps source fields → target fields Runs transform logic
Spreadsheet CSV / XLSX data Headers + records	Validation Mandatory fields Data type checks Duplicate checks	Target Table Creates new records Updates matched records
Error / Log Store Rejected rows Validation messages Import log	Import Result Success / failure counts Error details	User Dashboard Import status Summary and downloadable errors

1. Upload User uploads spreadsheet	2. Read System reads headers and rows	3. Map Transform Map connects fields	4. Transform Rules clean/convert values	5. Validate Checks data and duplicates	6. Import Insert/update target records	7. Report Shows import summary
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DFD Explanation

External Entity: Data Administrator/User uploads the spreadsheet and initiates the import. **Processes:** File reading, mapping, transformation, validation, and import. **Data Stores:** Import logs, error records, and the target table. **Outputs:** Imported records and an import summary showing successful, updated, and failed rows.

2. User Stories

The following user stories define the main functional requirements for spreadsheet import using Transform Maps.

User Type	Functional Requirement	User Story	Acceptance Criteria	Priority	Release
Data Administrator	Upload Spreadsheet	As a data administrator, I want to upload a CSV/XLSX file so that its records can be imported into the system.	Valid file is accepted; unsupported/empty file shows a clear error.	High	Sprint-1
Data Administrator	Select Target Table	As a data administrator, I want to select the target table so that the imported data reaches the correct destination.	Target table can be selected and displayed before mapping.	High	Sprint-1
Data Administrator	Create Transform Map	As a data administrator, I want to define source-to-target field mappings so that spreadsheet columns populate the correct fields.	Source and target fields are visible; mappings can be created and saved.	High	Sprint-1
Data Administrator	Transform Data	As a data administrator, I want transformation rules to run during import so that values are converted into the required format.	Configured transformation logic executes for applicable rows.	Medium	Sprint-2
Data Administrator	Validate Records	As a data administrator, I want the system to validate imported values so that invalid records are identified before completion.	Mandatory fields, formats, and relevant validation rules are checked.	High	Sprint-2
Data Administrator	Insert / Update Records	As a data administrator, I want matching records to be updated and new records inserted so that the target table remains current.	New rows create records; matched rows update according to the configured coalesce/lookup rule.	High	Sprint-2
Data Administrator	View Import Results	As a data administrator, I want to view the import summary so that I can confirm successful and failed rows.	Result shows total, successful, updated, and failed counts with error details.	Medium	Sprint-2
System Administrator	Review Import Logs	As a system administrator, I want to review import logs so that I can troubleshoot mapping or data errors.	Import execution details and error messages are retained for review.	Low	Sprint-3

3. Import Process – Step-by-Step

Step 1 – Prepare Spreadsheet

Create a spreadsheet with column headers matching the required source data. Remove blank rows and ensure values use the expected formats.

Step 2 – Upload the File

The user uploads the CSV/XLSX file through the import interface. The system checks file type and basic file validity.

Step 3 – Configure Import Set

The uploaded data is loaded into an import set/staging structure so that the source records can be processed safely before reaching the target table.

Step 4 – Configure Transform Map

Create source-to-target mappings. Define field maps, lookup/coalesce behavior, and transformation logic where required.

Step 5 – Transform and Validate

The system transforms values and validates mandatory fields, formats, references, and duplicate/matching conditions.

Step 6 – Import to Target

Valid records are inserted or updated in the target table according to the configured mapping and matching rules.

Step 7 – Review Results

The user reviews the import summary and investigates failed rows using the recorded error messages and import logs.

4. Functional Requirements

FR-01	The system shall accept a supported spreadsheet format such as CSV/XLSX.
FR-02	The system shall read source columns and records from the uploaded file.
FR-03	The system shall allow source-to-target field mapping through a Transform Map.
FR-04	The system shall apply configured transformation rules during processing.
FR-05	The system shall validate records and identify failed rows.
FR-06	The system shall insert new records and update matched records according to configured rules.
FR-07	The system shall record import status and error information.
FR-08	The system shall display a final import summary to the user.

5. Expected Outcome

After the workflow completes, valid spreadsheet data is available in the target table, matching records are updated according to the configured rules, invalid rows are identified with useful error information, and the administrator can verify the overall import status.

6. Conclusion

Transform Maps provide a controlled way to move spreadsheet data into a target system. By separating upload, mapping, transformation, validation, and import stages, the project improves consistency, traceability, and the ability to

identify data-quality issues.